

WASP-50b

R-0655

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_251231 · created 2026-07-27T17:12:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461040.72 +/- 0.12 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.03 +/- 1.78 [%]
Orbital Inclination (inc)	85.37 +/- 2.78
Airmass coefficient 1 (a1)	1.009 +/- 0.032
Airmass coefficient 2 (a2)	-0.002 +/- 0.013
Scatter in the residuals of the lightcurve fit is	3.16 %
Best Comparison Star	#1 - [221, 385]
Optimal Method	PSF photometry
Transit Duration (day)	0.034 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.174 ± 0.051 vs 0.1404 (deviation 0.66σ)
Duration fit vs published	0.034 d vs 0.0754 d (deviation 1.22σ)
Mid-time O–C	38.7 min
Depth SNR	3.4
Residual scatter	3.16 %

Light curve

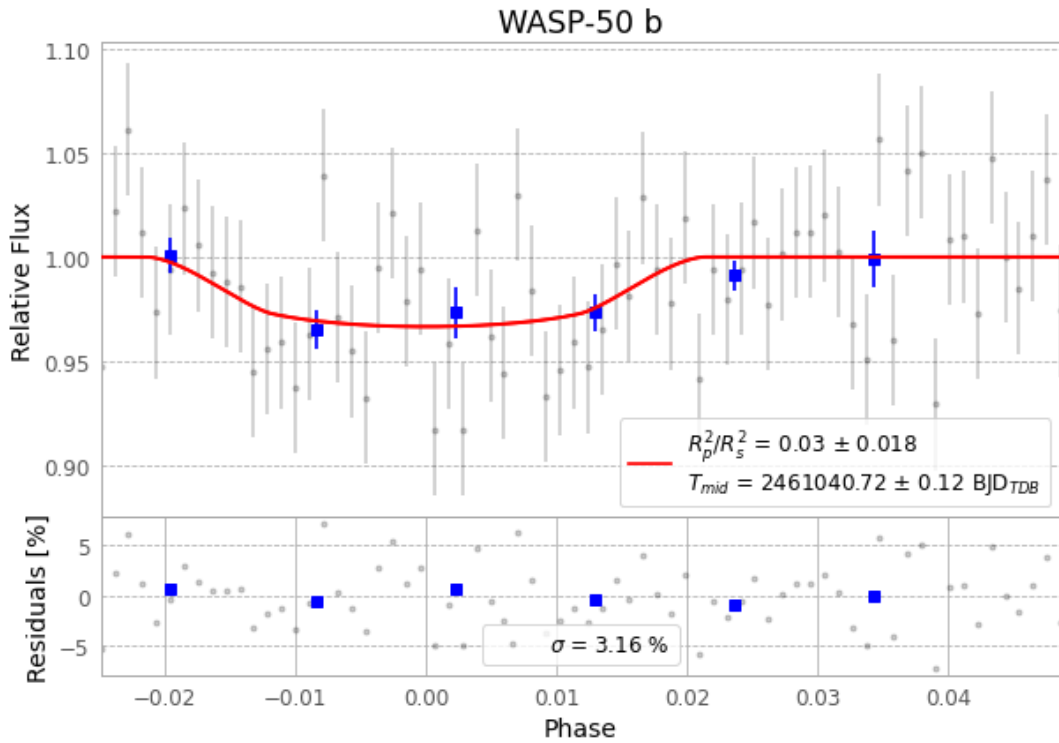


Figure 1 — FinalLightCurve_WASP-50 b_2025-12-30.png (SHA-256 50d82dcc63baec22...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [576, 345], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eda3f8a0fa2b01ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

70 FITS frames (46 MB), WASP-50251231040316.FITS → WASP-50251231073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-251231.log (183f8373f002...), FinalLightCurve_WASP-50 b_2025-12-30.png (50d82dcc63ba...), FinalLightCurve_WASP-50 b_2025-12-30.csv (139c2ab9b15f...)

Compute resources

Wall time 205.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:13Z.